Zahra Helalizadeh

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Major in Digital Systems
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GitHub Profile
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Research Interests

- **Data Structures & Algorithms**
- **Microprocessor Design & Architecture**
- **ASIC/FPGA Design**
- **VLSI Design**

Education

•Sharif University of Technology *Ongoing*

Bachelor's Degree in Electrical Engineering, Major in Digital Systems Cumulative Average: 16.32/20

•Farzanegan's School of Development of Exceptional Talents *2018 – 2021*

Post Graduate Diploma in Mathematics and Physics Cumulative Average: 19.45/20

Projects

•Hardware Optimization for Large Language Models *Ongoing*

Hardware Optimizer

– Developing advanced techniques to optimize hardware for large language models on systems with limited memory capacity. Focused on efficient parameter storage and dynamic memory management to enable execution of models larger than available memory. The project aims to enhance inference speed and resource utilization on constrained systems.

•Design & Implementation of Field Programmable Gate Array *Ongoing*

Hardware Designer

– Designing and implementing FPGA chip layouts, focusing on key components such as configurable logic blocks, connection blocks, and switch boxes. Responsible for optimizing performance, resource utilization, and routing efficiency in line with technical specifications. Also exploring innovative digital design techniques, power efficiency strategies, and dynamic reconfiguration to enhance the overall FPGA architecture.

Course Projects

•Parallel Algorithm Design & Programming *Spring 2024*

CUDA Developer

– Enhanced expertise in high-performance computing by designing and implementing parallel algorithms using CUDA in the Parallel Architecture and Programming course. Focused on optimizing computational efficiency and developing scalable solutions for data-intensive tasks.

•Signal Processing on FPGA Systems *Spring 2024*

Digital Signal Processing Contributor

– Completed a complex FPGA/ASIC system design project, generating and processing signals on a Zynq platform. Gained comprehensive knowledge of digital communication protocols and signal processing techniques.

•Performance Optimization Algorithms & Programming *Spring 2024*

Microprocessor Systems Contributor

– Optimized matrix multiplication code in C for the Design of Microprocessor Systems course, employing advanced techniques such as pointer manipulation, cache optimization, and parallel processing with OpenMP to enhance performance.

•Digital Signal Analysis *Spring 2023*

Data Analyst

– Applied advanced MATLAB techniques in EEG signal analysis during the Signals and Systems course. Extracted key insights from EEG data, aiding in the understanding of brain activity and identifying unique brainwave patterns.

•Web Development & Application Programming *Spring 2022*

Full-Stack Developer

– Developed full-stack web applications in the Object-Oriented Programming course, creating responsive interfaces with HTML and CSS. Built a Java-based messaging application with SQL integration for efficient data handling and user communication.

•Game Development & Graphical Programming *Fall 2021*

C++ and SDL Developer

– Mastered game development in C++ during the Programming Fundamentals course, utilizing SDL for multimedia handling. Demonstrated proficiency in creating interactive gaming experiences by integrating audiovisual components with OpenGL and Direct3D.

Experience

•Real-Time Video Streaming Development *Summer 2024*

Linux Developer

– Developed a comprehensive RTSP streaming solution using FFmpeg, GStreamer, VLC, and OpenCV. Established an RTSP server with MediaMTX, facilitating real-time streaming of webcam feeds. Optimized streaming pipelines to enhance video quality and minimize latency, significantly improving real-time video communication.

•Web Development & Front-End Programming *Fall 2023*

Front-End Developer

– Contributed to the technical team for ReACT 2024, developing a dynamic website using Django, Python, and JavaScript. Implemented interactive features to enhance user engagement, playing a key role in the conference's success with over 500 international participants.

Relevant Courses

•Data Analytics & Intelligent Systems

– Machine Learning: Ongoing *Spring 2025*
– Artificial Intelligence: Ongoing *Fall 2024*

•Digital Systems & Hardware Design

– VLSI Design†: Ongoing *Fall 2024*
– Computer Interface Circuits†: Ongoing *Fall 2024*
– Advanced Copmuter Architecture†: Ongoing *Fall 2024*
– ASIC/FPGA Systems Lab: 19.1/20 *Spring 2024*
– ASIC/FPGA System Design: 15.8/20 *Spring 2024*
– Microprocessor Systems Design: 17/20 *Spring 2024*
– Computer Architecture & Microprocessors & Lab: 17.2/20 *Spring 2023*
– Logical Circuits & Digital Systems & Lab: 16.5/20 *Fall 2022*

•Algorithms & Programming

– Parallel Programming & Architecture†: 16.2/20 *Spring 2024*
– Data Structures & Algorithms: 18.5/20 *Spring 2024*
– Object Oriented Programming: 20/20 *Spring 2022*
– Fundamentals of Programming: 19.6/20 *Fall 2021*

Positions of Responsibility

•Computer Architecture & Microprocessor Course *Fall 2023, Spring 2024, Fall 2024*

Teaching Assistant

– Designed and graded homework assignments, as well as midterm and final exams. Provided guidance to students in understanding complex architectural concepts and assisted with problem-solving during office hours.

•Logical Circuits & Digital Systems Course *Fall 2023, Fall 2024*

Teaching Assistant

– Designed and graded homework assignments and quizzes. Facilitated learning by answering student inquiries and helping them grasp the fundamental principles of logical circuits.

•Basic Programming Course *Fall 2024*

Teaching Assistant

– Designed quizzes and exams, graded them, and assisted students in understanding key programming concepts. Addressed student questions and supported their progress in mastering basic programming skills.

•Engineering Mathematics Course *Fall 2023*

Teaching Assistant

– Graded homework assignments, providing constructive feedback to enhance student understanding of mathematical concepts. Actively engaged with students by answering questions and offering additional support to clarify challenging topics.

Skills

•Programming Languages: C/C++, CUDA, Java, Python, MATLAB

•Web Development: HTML, CSS, JavaScript, Django

•Database Management: mySQL, PostgreSQL

•Hardware Description Languages: VHDL, Verilog

•Operating Systems: Windows, Linux

•Technical Skills: Microsoft Office, LaTeX

•Languages: Advanced English, Native Persian

†: Graduate Course